

15 DAYS OF LEARNING & GROWTH

ICT ACADEMY TRAINING

Technical Skills • Soft Skills • Software Testing • Project Development

STUDENT NAME

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DEPARTMENT

CSE / ISE

COLLEGE

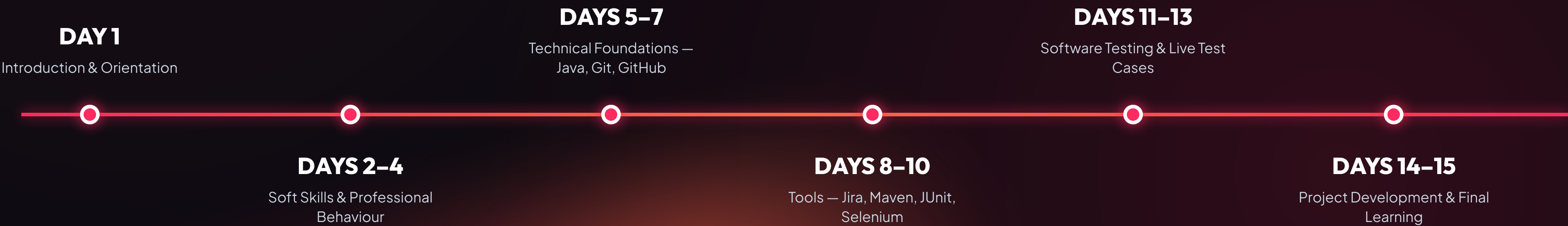
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DURATION

15 Days — ICT Academy

OUR 15-DAY LEARNING JOURNEY

A structured program that combined learning + practice + teamwork + project implementation — covering both technical development and professional growth.



WHAT WE **LEARNED**

Comprehensive 4-pillar foundation combining behavioral excellence, modern programming, industry testing standards, and agile project delivery.



SOFT SKILLS

- Communication & workplace etiquette
- Professional behaviour & body language
- Greetings & executive self-introductions
- Respecting opinions & team active listening



TECHNICAL SKILLS

- Java OOP & Full-Stack Architecture
- Git & GitHub Distributed Version Control
- Jira Agile Tracking & Google Stitch IDE
- Node.js, Express 5, React 19 Ecosystem



SOFTWARE TESTING

- Testing fundamentals & QA life cycle
- Manual & automated browser testing
- Selenium WebDriver automation scripts
- JUnit & Jest unit test cases with assertions



PROJECT DEV

- Team collaboration & sprint workflows
- Dual database integration (SQLite & Supabase)
- End-to-end testing & payment verification
- Real-world full-stack cloud deployment

SOFT SKILLS & PROFESSIONAL COMMUNICATION

Building the emotional intelligence, executive presence, and collaborative mindset required in corporate software engineering environments.

WHAT WE DEVELOPED

→ GREETINGS & INTRODUCTIONS

Professional ways to meet, greet, and introduce ourselves in corporate meetings and workplace environments with impact.

→ CONFIDENT COMMUNICATION

Speaking clearly, active listening during technical reviews, asking insightful questions, and expressing ideas with poise.

→ WORKPLACE ETIQUETTE

Respecting diverse perspectives, maintaining professional email decorum, and resolving conflicts collaboratively as a team.

THE TRANSFORMATION

BEFORE

Hesitation, informal unstructured communication, fear of speaking in groups, and limited presentation confidence.

AFTER

Articulate clarity, professional self-assurance, active constructive listening, and structured team problem-solving.

The ICT Academy training provided us with actionable tools to communicate respectfully and lead effectively in any modern technical team.

TECHNICAL SKILLS & TOOLS WE EXPLORED

Industry-standard tools and frameworks that shaped our full-stack engineering, testing, and deployment workflows.



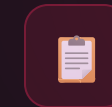
JAVA & DEV FOUNDATIONS

OOP concepts, modular code architecture, exception handling, and integrating with Maven & JUnit for test suites.



GIT & GITHUB

Distributed version control, branch management, pull requests, resolving merge conflicts, and team code reviews.



JIRA & GOOGLE STITCH

Agile sprint boards, user story backlog grooming, assigning responsibilities, defect logging, and AI-accelerated workflows.



MAVEN, JUNIT & SELENIUM

Build lifecycle automation, dependency management, unit testing assertions, and browser-driven UI test automation.



ANTIGRAVITY IDE

Full-stack development workspace, real-time code generation, intelligent refactoring, and debugging across repositories.



DEPLOYMENT & CLOUD

End-to-end release engineering: AWS (S3, CloudFront, EC2), Vercel serverless hosting, Docker containerization, and HTTPS.

SELENIUM & SOFTWARE TESTING LIFECYCLE

Testing is an indispensable pillar of software engineering. We gained practical mastery across manual testing, automated script execution, and defect lifecycles.

1 MANUAL TESTING CONCEPTS

Writing comprehensive test scenarios, mapping boundary value conditions, verifying expected vs. actual results, and bug logging.

2 AUTOMATION TESTING WITH SELENIUM

Writing and executing WebDriver automation scripts in Java to test end-to-end browser interactions, form validations, and routing.

3 JUNIT & API UNIT TESTING

Component-level unit tests, verifying pass/fail assertions, and automated REST endpoint validation with Jest & Supertest.

4 LIVE TEST CASES & DEFECT TRACKING

Executing live regression and functional test cases on our Backlox project application to ensure zero production glitches.

TESTING LIFECYCLE



OUR MAJOR PROJECT: **BACKLOX**

CAREER GUIDANCE PLATFORM FOR 12TH-PASS STUDENTS — A unified roadmap & mentoring web application empowering students with curated guidance across engineering, medicine, and technology streams.

THE PROBLEM

High school students completing 12th grade face immense confusion regarding career pathways, branch specializations, essential semester skills, and real-world IT industry role requirements.

OUR SOLUTION

BACKLOX — An interactive, full-stack career platform designed to help students explore post-12th streams, compare branch curriculum, follow semester roadmaps, build project portfolios, and prepare for high-demand jobs.

WHAT WE BUILT

A high-performance SPA featuring branch roadmaps (CSE, IT, AI/DS, ECE, Mech, Civil), student dashboards, project recommendations, AI-assisted career advisory, and a secure Razorpay Pro unlock.

BACKLOX: TECHNOLOGY STACK & DATABASES

A robust, scalable full-stack ecosystem combining modern reactive frontend interfaces, dual-database persistence, automated testing, and cloud infrastructure.

FRONTEND **React 19 & Vite 8**

SPA architecture, React Router v7, Context API state management, responsive dark-mode Glassmorphism CSS3, and Oxlint code quality.

BACKEND **Node.js & Express 5**

Modular RESTful API routes, JWT bearer authentication, Bcrypt.js password security, Multer asset handling, and custom RAG AI query engine.

DATABASES **SQLite & PostgreSQL (Supabase)**

Dual data storage: **better-sqlite3** for fast local transactional persistence + **Supabase Cloud SQL** with Row-Level Security (RLS) & serverless JSON fallback.

PAYMENTS & DEVOPS **Razorpay & AWS Cloud**

Razorpay HMAC-SHA256 signature verification, AWS S3/CloudFront CDN/EC2 deployment, Vercel edge routing, and Docker containerization.

PROJECT LIFECYCLE & WORKFLOW



PLANNING
Jira Sprints & User Stories



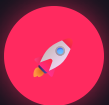
DEVELOP
React, Node, Express, DB



VERSION
Git & GitHub PR Workflow



TESTING
Selenium, Jest & Supertest



DEPLOY
AWS & Vercel Production

“A successful project is not just about writing code — it is about communication, database architecture, rigorous testing, and end-to-end team responsibility.”

WHAT WE GAINED **IN 15 DAYS**

A fast-paced journey from academic conceptual theory to industry-grade full-stack implementation, quality assurance, and professional teamwork.

BEFORE

Limited exposure to industry standard tools, lack of professional QA testing workflows, and hesitation in agile collaboration.

AFTER

Hands-on mastery of full-stack engineering, test-driven methodologies (Selenium/Jest), database design, and confident presentation skills.

TECHNICAL TAKEAWAYS

Testing is mandatory, not optional. Scalable architecture and version control enable seamless collaborative delivery.

15 DAYS

Structured high-intensity learning

NEW TOOLS

Modern full-stack & testing stack

SOFT SKILL TAKEAWAYS

Active listening builds stronger teams. Clear communication and accountability transform coders into engineers.

PRACTICE

Applied live test cases & code

GROWTH

Career readiness & teamwork

THANK YOU

15 DAYS OF LEARNING. COUNTLESS LESSONS. ONE STEP FORWARD.

The ICT Academy training was not just about learning individual technologies — it was about understanding how to think, communicate, test, and engineer software like industry professionals.

TRAINING

ICT Academy — 15-Day Industry Readiness Program

EXPERIENCE

Hands-on • Practical • Collaborative • Quality-Driven

PROJECT

BACKLOX — Full-Stack Post-12th Career Guidance Platform

| **QUESTIONS & DISCUSSION**